EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 16

Basic Maths and Calculus

Definite Integrals, Area and Substitution

By - Excellence Programme

23 September 2026

Topics to be covered

01 Indefinite-integral recap

02 Definite integral: $\int f(x)\,dx$ from a to b

03 Fundamental rule: $F(b) - F(a)$

04 No +C in definite integrals

05 Area under a curve

06 Definite practice · then substitution (u, du)

07 Displacement = area under v–t · Substitution

Indefinite-integral recap

$$\int f(x)\,dx = F(x) + C$$

$$\int x^n\,dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$$

$$\int \sin x\,dx = -\cos x + C \quad \cdot \quad \int \cos x\,dx = \sin x + C$$

Check by differentiating: you must recover f(x)

Quick check: indefinite

Find $\int (3x^2 - 4)\,dx$.

Solution: Quick check: indefinite

Step 1: $\int 3x^2\,dx = x^3$

Step 2: $\int -4\,dx = -4x$

$x^3 - 4x + C$.

Rough work / extra work

Definite integral

A definite integral has lower and upper limits. The answer is a number.

$$\int f(x)\,dx \text{ from } a \text{ to } b = [F(x)]_a^b = F(b) - F(a)$$

F is any antiderivative of f. The constant C cancels, so we omit +C.

Never write +C in a definite integral.

Three easy steps

1. Find the antiderivative F(x) of f(x)

2. Put the upper limit: F(b)

3. Subtract the lower limit: F(b) − F(a)

Answer = F(upper) − F(lower)

First definite integral

Evaluate $\int 2x\,dx$ from 0 to 2.

Solution: First definite integral

Step 1: Antiderivative: $F(x) = x^2$

Step 2: $F(2) - F(0) = 4 - 0$

$\int 2x\,dx$ from 0 to 2 = 4.

Power definite integral

Evaluate $\int x^2\,dx$ from 1 to 3.

Solution: Power definite integral

Step 1: $F(x) = x^3/3$

Step 2: $F(3) - F(1) = 27/3 - 1/3 = 9 - 1/3$

26/3.

Rough work / extra work

Polynomial definite

Evaluate $\int (4x^3 - 2x)\,dx$ from 0 to 1.

Solution: Polynomial definite

Step 1: $F(x) = x^4 - x^2$

Step 2: $F(1) - F(0) = (1 - 1) - 0$

0.

Trig definite

Evaluate $\int \cos x \, dx$ from 0 to $\pi/2$.

Solution: Trig definite

Step 1: $F(x) = \sin x$

Step 2: $F(\pi/2) - F(0) = \sin(\pi/2) - \sin 0 = 1 - 0$

1.

Rough work / extra work

Sine definite

Evaluate $\int \sin x \, dx$ from 0 to $\pi/2$.

Solution: Sine definite

Step 1: $F(x) = -\cos x$

Step 2: $F(\pi/2) - F(0) = (-\cos \pi/2) - (-\cos 0) = 0 - (-1)$

1.

Area under a curve

When f(x) ≥ 0 on [a, b], the definite integral equals the area between the curve and the x-axis.

$$\text{Area} = \int f(x)\,dx \quad (f \geq 0) \text{ from } a \text{ to } b$$

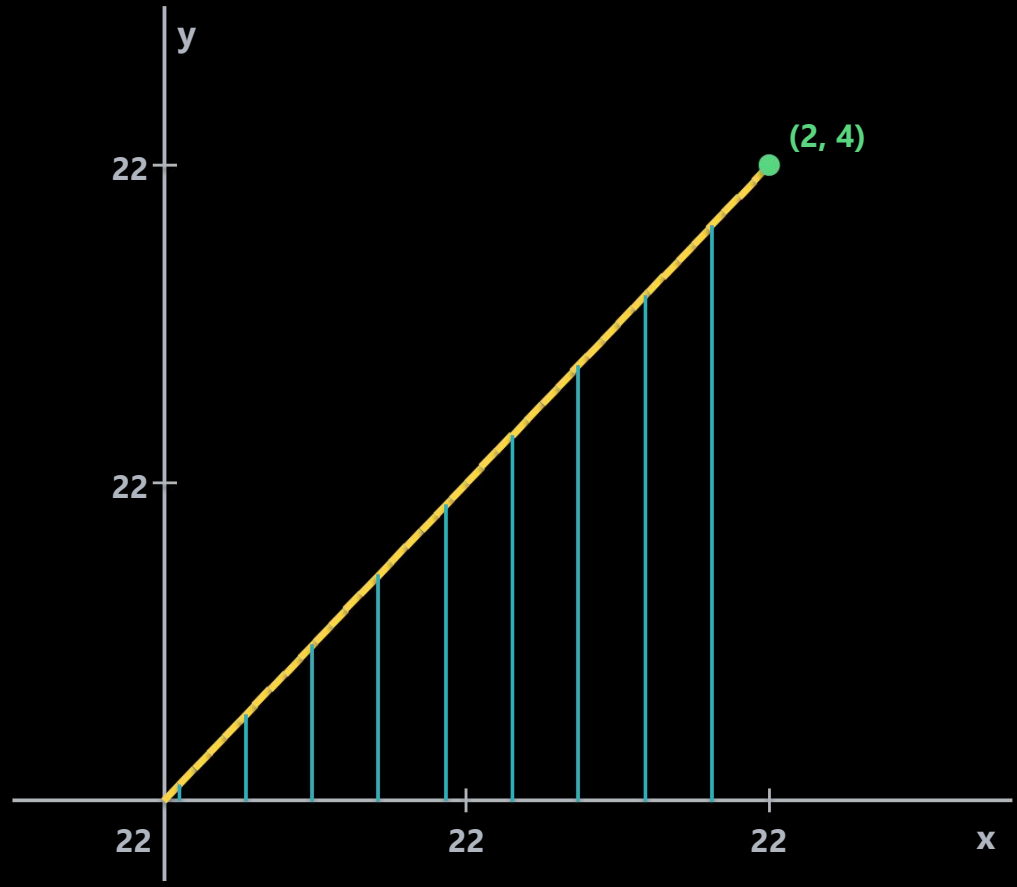
If the curve dips below the axis, the integral counts that region as negative.

Area of a region below the axis needs absolute value (or split the integral).

Area under y = 2x from 0 to 2

The region is a right triangle of base 2 and height 4.

Area = ½ · 2 · 4 = 4 = ∫ 2x dx from 0 to 2



Area under a line

Find the area under $y = x$ from $x = 0$ to $x = 4$.

Solution: Area under a line

Step 1: Area = $\int x\,dx$ from 0 to 4 = $[x^2/2]$ from 0 to 4

Step 2: = 16/2 − 0 = 8

Area = 8 square units.

Rough work / extra work

Area under a parabola

Find the area under $y = x^2$ from $x = 0$ to $x = 2$.

Solution: Area under a parabola

Step 1: Area = $\int x^2\,dx$ from 0 to 2 = $[x^3/3]$ from 0 to 2

Step 2: = 8/3 − 0

Area = 8/3 square units.

Area under cosine

Find the area under $y = \cos x$ from $x = 0$ to $x = \pi/2$.

Solution: Area under cosine

Step 1: Area = $\int \cos x \, dx$ from 0 to $\pi/2$ = $[\sin x]$ from 0 to $\pi/2$

Step 2: = 1 − 0 = 1

Area = 1 square unit.

Rough work / extra work

Two useful properties

$$\int f \text{ from } a \text{ to } b = -\int f \text{ from } b \text{ to } a$$

$$\int f \text{ from } a \text{ to } b = \int_a^c f + \int_c^b f$$

You may split an interval at a convenient point c.

Swapping limits changes the sign of the integral.

Swap limits

If $\int 2x\,dx$ from 0 to 2 = 4, find $\int 2x\,dx$ from 2 to 0.

Solution: Swap limits

Step 1: Swapping limits multiplies by −1

$\int 2x\,dx$ from 2 to 0 = −4.

Split the interval

Evaluate $\int x\,dx$ from 0 to 4 by splitting at $x = 2$.

Solution: Split the interval

Step 1: $\int = \int + \int$ from 2 to 4 from 0 to 4 from 0 to 2

Step 2: $\int x\,dx$ from 0 to 2 $= 2 \cdot \int x\,dx$ from 2 to 4 $= [x^2/2]$ from 2 to 4 $= 8 - 2 = 6$

2 + 6 = 8.

Rough work / extra work

Constant integrand

Evaluate $\int 3\,dx$ from 1 to 5.

Solution: Constant integrand

Step 1: $F(x) = 3x$

Step 2: $F(5) - F(1) = 15 - 3 = 12$

12.

Exponential definite

Evaluate ∫ e^x dx from 0 to 1.

Solution: Exponential definite

Step 1: F(x) = e^x

Step 2: F(1) − F(0) = e − 1

e − 1.

Mixed polynomial

Evaluate $\int (x^3 + 2x)\,dx$ from 0 to 2.

Solution: Mixed polynomial

Step 1: $F(x) = x^4/4 + x^2$

Step 2: $F(2) - F(0) = (4 + 4) - 0 = 8$

8.

Rough work / extra work

Physics: area under v–t graph

On a velocity–time graph, the area between the curve and the t-axis is displacement.

$$\text{displacement} = \int v(t)\,dt \text{ from } t_1 \text{ to } t_2 = \text{area under v–t}$$

Same idea as last class: integrating velocity recovers position change.

Area below the t-axis means displacement in the opposite direction.

Constant velocity displacement

If $v = 5\ m\ s^{-1}$ from $t = 0$ to $t = 4\ s$, find the displacement.

Solution: Constant velocity displacement

Step 1: $s = \int 5\,dt$ from 0 to 4 = [5t] from 0 to 4 = 20 − 0

Step 2: Area of rectangle = 5 Ã— 4 = 20

Displacement = 20 m.

Linear velocity displacement

If $v = 2t$ (m s^{-1}) from $t = 0$ to $t = 3$ s, find the displacement.

Solution: Linear velocity displacement

Step 1: $s = \int 2t\,dt$ from 0 to 3 = $[t^2]$ from 0 to 3 = 9

Step 2: Triangle area = ½ · 3 · 6 = 9

Displacement = 9 m.

Rough work / extra work

From a to Δv

If $a = 3\ m\ s^{-2}$ (constant) from $t = 0$ to $t = 2\ s$ and $u = 0$, find v at $t = 2$.

Solution: From a to Δv

Step 1: $\Delta v = \int a\,dt$ from 0 to 2 $= \int 3\,dt$ from 0 to 2 $= 6$

Step 2: $v = u + \Delta v = 0 + 6$

$v = 6\ \text{m s}^{-1}$.

Homework: definite power

Evaluate $\int (3x^2 - 1)\,dx$ from 1 to 2.

Solution: Homework: definite power

Step 1: $F(x) = x^3 - x$

Step 2: $F(2) - F(1) = (8 - 2) - (1 - 1) = 6$

6.

Rough work / extra work

Homework: area under line

Find the area under $y = 3x$ from $x = 1$ to $x = 3$.

Solution: Homework: area under line

Step 1: Area = $\int 3x\,dx$ from 1 to 3 = $[3x^2/2]$ from 1 to 3

Step 2: = (27/2) − (3/2) = 24/2 = 12

Area = 12 square units.

Homework: check by graph

v falls from 10 m/s to 0 in 5 s on a straight v–t graph. Find displacement.

Solution: Homework: check by graph

Step 1: Triangle area = ½ · 5 · 10 = 25

Step 2: Or: $v = 10 - 2t$; $\int (10 - 2t)\,dt$ from 0 to 5 = $[10t - t^2]$ from 0 to 5 = 50 − 25

Displacement = 25 m.

Definite-integral lock lines

∫ f(x) dx from a to b = F(b) − F(a) (no +C)

If f ≥ 0, this equals area under the curve from a to b

∫ f from a to b = −∫ f from b to a · split intervals when useful

Displacement = area under the v–t graph

Rough work / extra work

Practice

1. Redo every definite-integral question without the solution.

2. For each area question, also sketch the region.

3. Convert one v–t triangle or rectangle into an ∫ statement.

4. Write: indefinite needs +C; definite does not.

Forgetting to subtract F(a) is the most common definite-integral error.

Quick recap, then substitution

$\int f(x)\,dx$ from a to b = $F(b) - F(a)$ (no $+C$)

If $f \geq 0$, this is the area under the curve

Displacement = area under the v–t graph

$\int x^n\,dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$

Quick check: definite

Evaluate $\int 2x\,dx$ from 0 to 3.

Solution: Quick check: definite

Step 1: $F(x) = x^2$

Step 2: $F(3) - F(0) = 9 - 0$

9.

Rough work / extra work

Substitution: chain rule in reverse

Chain rule: d/dx [F(u)] = F'(u) · du/dx.

So if the integrand looks like f(u) · (du/dx), integrate with respect to u.

$$\int f(u)\,(du/dx)\,dx = \int f(u)\,du$$

Always replace dx as well — do not leave a mixture of u and x.

Four board steps

1. Choose u = (the inside function)

2. Differentiate: du/dx = … → write du = (…) dx

3. Rewrite the whole integral in u only

4. Integrate, then put x back (or change limits)

Simple linear inside

Find $\int (2x + 3)^3 \cdot 2\,dx$.

Solution: Simple linear inside

Step 1: Let $u = 2x + 3 \rightarrow du = 2\,dx$

Step 2: Integral becomes $\int u^3\,du = u^4/4 + C$

$(2x + 3)^4/4 + C$.

Missing constant factor

Find $\int (2x + 3)^3 \, dx$.

Solution: Missing constant factor

Step 1: $u = 2x + 3 \rightarrow du = 2\,dx \rightarrow dx = du/2$

Step 2: $\int u^3 (du/2) = (1/2) \cdot u^4/4 + C = u^4/8 + C$

$(2x + 3)^4/8 + C.$

If du has a constant factor, bring 1/that factor outside.

Rough work / extra work

Power of a linear expression

Find $\int \sqrt{(3x + 1)}\, dx$.

Solution: Power of a linear expression

Step 1: $u = 3x + 1 \rightarrow du = 3\,dx \rightarrow dx = du/3$

Step 2: $\int u^{1/2}\,(du/3) = (1/3)\cdot u^{3/2}/(3/2) + C$

Step 3: $= (1/3)\cdot(2/3)\,u^{3/2} + C = (2/9)\,u^{3/2} + C$

$(2/9)(3x + 1)^{3/2} + C.$

Exponential substitution

Find $\int e^{2x} \cdot 2\,dx$.

Solution: Exponential substitution

Step 1: $u = 2x \rightarrow du = 2\,dx$

Step 2: $\int e^u\,du = e^u + C$

$e^{(2x)} + C$.

Rough work / extra work

Exponential without the 2

Find ∫ e^(2x) dx.

Solution: Exponential without the 2

Step 1: $u = 2x \rightarrow dx = du/2$

Step 2: $\int e^u (du/2) = (1/2) e^u + C$

$(1/2) e^{(2x)} + C$.

Sine of a linear angle

Find $\int \sin(3x)\,dx$.

Solution: Sine of a linear angle

Step 1: $u = 3x \rightarrow dx = du/3$

Step 2: $\int \sin u \, (du/3) = -(1/3)\cos u + C$

$-(1/3)\cos(3x) + C$.

Cosine of a linear angle

Find $\int \cos(5x)\,dx$.

Solution: Cosine of a linear angle

Step 1: $u = 5x \rightarrow dx = du/5$

$(1/5)\sin(5x) + C.$

Rough work / extra work

Chain-rule lookalike

Find $\int 2x \cos(x^2)\, dx$.

Solution: Chain-rule lookalike

Step 1: $u = x^2 \;\rightarrow\; du = 2x\,dx$

Step 2: $\int \cos u\,du = \sin u + C$

$\sin(x^2) + C$.

Reciprocal substitution

Find $\int (2x)/(x^2 + 1)\, dx$.

Solution: Reciprocal substitution

Step 1: $u = x^2 + 1 \rightarrow du = 2x\,dx$

Step 2: $\int du/u = \ln|u| + C$

$\ln|x^2 + 1| + C$.

Rough work / extra work

Definite integrals with substitution

Two clean options: change the limits to u, or substitute back to x then use $F(b) - F(a)$.

Preferred: change limits with u → no need to return to x

If you change variable, you must also change both limits.

Changing limits

When x = a, compute u(a)

When x = b, compute u(b)

Then evaluate ∫ f(u) du from u(a) to u(b)

Do not put x-limits on a u-integral.

Definite with linear u

Evaluate $\int (2x + 1)^3$ from 0 to $1 \cdot 2$ dx.

Solution: Definite with linear u

Step 1: $u = 2x + 1 \rightarrow du = 2\,dx$

Step 2: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 3$

Step 3: $\int u^3\,du$ from 1 to 3 $= [u^4/4]$ from 1 to 3 $= 81/4 - 1/4$

20.

Definite sine substitution

Evaluate $\int \cos(3x)$ from 0 to $\pi/6 \cdot 3\,dx$.

Solution: Definite sine substitution

Step 1: $u = 3x \rightarrow du = 3\,dx$

Step 2: $x = 0 \rightarrow u = 0$; $x = \pi/6 \rightarrow u = \pi/2$

Step 3: $\int \cos u\,du$ from 0 to $\pi/2 = [\sin u]$ from 0 to $\pi/2 = 1$

1.

Rough work / extra work

Definite without the factor ready

Evaluate ∫ e^(2x) dx from 0 to 1.

Solution: Definite without the factor ready

Step 1: $u = 2x \rightarrow dx = du/2$; $x = 0 \rightarrow u = 0$; $x = 1 \rightarrow u = 2$

Step 2: $\int e^u (du/2)$ from 0 to 2 $= (1/2)[e^u]$ from 0 to 2 $= (1/2)(e^2 - 1)$

$(e^2 - 1)/2$.

Definite reciprocal form

Evaluate $\int 2x/(x^2 + 1)\,dx$ from 0 to 1.

Solution: Definite reciprocal form

Step 1: $u = x^2 + 1 \rightarrow du = 2x\,dx$

Step 2: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$

Step 3: $\int du/u$ from 1 to 2 $= [\ln|u|]$ from 1 to 2 $= \ln 2 - \ln 1$

$\ln 2$.

Rough work / extra work

Physics with substitution

When acceleration or velocity is a composite function of time, substitute the inside.

Same four steps — then put the physical limits of t

Keep units on the final number.

Velocity from a = 2t + 1

If a = 2t + 1 and v = 0 at t = 0, find v at t = 2.

Solution: Velocity from a = 2t + 1

Step 1: $v = \int (2t + 1)\,dt$ from 0 to 2 $= [t^2 + t]$ from 0 to 2

Step 2: $= (4 + 2) - 0 = 6$

$v = 6\ m\ s^{-1}$.

Displacement with $v = 3t^2$

If $v = 3t^2$ and $x = 0$ at $t = 0$, find x at $t = 2$.

Solution: Displacement with $v = 3t^2$

Step 1: $x = \int 3t^2\,dt$ from 0 to 2 $= [t^3]$ from 0 to 2 $= 8$

$x = 8$ m.

Rough work / extra work

Composite velocity

If $v = 2t\cos(t^2)$ and $x = 0$ at $t = 0$, find x at $t = \sqrt{(\pi/2)}$.

Solution: Composite velocity

Step 1: $u = t^2 \rightarrow du = 2t\,dt$; $t = 0 \rightarrow u = 0$; $t = \sqrt{(\pi/2)} \rightarrow u = \pi/2$

Step 2: $x = \int \cos u\,du$ from 0 to $\pi/2 = [\sin u]$ from 0 to $\pi/2 = 1$

$x = 1$ m.

Mixed: identify u

Find $\int 6x^2 (x^3 + 4)^5 \, dx$.

Solution: Mixed: identify u

Step 1: $u = x^3 + 4 \rightarrow du = 3x^2\,dx \rightarrow 6x^2\,dx = 2\,du$

Step 2: $\int 2u^5\,du = 2 \cdot u^6/6 + C = u^6/3 + C$

$(x^3 + 4)^6/3 + C.$

Mixed: definite power

Evaluate $\int 4x\,(x^2 + 1)^3\,dx$ from 0 to 1.

Solution: Mixed: definite power

Step 1: $u = x^2 + 1 \rightarrow du = 2x\,dx \rightarrow 4x\,dx = 2\,du$

Step 2: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$

Step 3: $\int 2u^3\,du$ from 1 to 2 $= 2[u^4/4]$ from 1 to 2 $= (1/2)(16 - 1)$

15/2.

Rough work / extra work

Homework: sin(2x)

Find $\int \sin(2x)\,dx$.

Solution: Homework: sin(2x)

Step 1: $u = 2x \rightarrow dx = du/2$

$-(1/2)\cos(2x) + C.$

Homework: definite e^(3x)

Evaluate ∫ e^(3x) dx from 0 to 1.

Solution: Homework: definite e^(3x)

Step 1: $u = 3x \rightarrow dx = du/3$; limits $0 \rightarrow 3$

Step 2: $(1/3)[e^u]$ from 0 to 3 = $(e^3 - 1)/3$

$(e^3 - 1)/3$.

Rough work / extra work

Homework: check by differentiate

Differentiate $y = (1/5)\sin(5x)$ and confirm it matches $\cos(5x)$.

Solution: Homework: check by differentiate

Step 1: $\frac{dy}{dx} = \frac{1}{5} \cdot \cos(5x) \cdot 5 = \cos(5x)$

Check successful — so $\int \cos(5x)\,dx = \frac{1}{5}\sin(5x) + C$.

Substitution lock lines

Substitution = chain rule written backwards

Choose u, write du, rewrite the whole integral

For definite integrals, change both limits with u

Always check by differentiating the answer

Practice

1. Redo every substitution question without looking.

2. For each indefinite answer, differentiate and recover the integrand.

3. Practise changing limits on three definite examples.

4. Write one physics $\int$ with a composite function of t.

Leaving a mix of u and x in the same integral is the classic trap.